Mechanistic Analysis and Resolution of the SolarLab–SCAPS-1D Discrepancies on the Partner Base Model

Abstract

This document identifies, verifies, and resolves the residual differences between the SolarLab device simulator and the reference solver SCAPS-1D on the partner Base Model. The open-circuit-voltage deficit — previously attributed first to quasi-Fermi-level dissipation across the heterojunction band offsets and then to the contact boundary condition, both of which are refuted here — is traced to a single, exactly quantified discretisation omission: the Scharfetter–Gummel flux carried the band-edge potentials ($\varphi + \chi$, $\varphi + \chi + E_g$) but not the **effective-density-of-states terms** ($V_T \ln N_C$, $-V_T \ln N_V$). The omission imposes a spurious quasi-Fermi-level step of $kT \ln(\text{DOS ratio})$ at every DOS-contrast heterojunction — 137 mV on this stack (83 meV at HTL/PVK, 54 meV at PVK/ETL), matching the measured quasi-Fermi-level profile exactly. With the correction implemented (`dos_band_potentials`), a refined voltage grid, and a $V_{oc} < V_{bi}$ physical-validity guard on the sweep pipeline, **all six parameter-sweep trends now match SCAPS in direction** — including the two historical direction mismatches (ETL donor doping and perovskite donor doping) — and the base figures of merit agree to −50 mV in V_{oc}, +0.9 pp in FF, and −2 % in J_{sc}. The remaining differences are individually named and bounded.

1. Discrepancy overview

Table 1 compares the base operating point in three configurations: the originally published SolarLab numbers, the corrected configuration, and SCAPS-1D.

Metric	SolarLab (published)	SolarLab (corrected)	SCAPS-1D	Residual
V_{oc} (V)	1.072	**1.118**	1.168	−50 mV
J_{sc} (mA/cm^2)	25.73	25.70	26.28	−2 %
FF (%)	85.6	**87.9**	87.0	+0.9 pp
PCE (%)	23.6	**25.26**	26.69	−1.4 pp

Table 1. Base operating-point figures of merit. “Corrected” = the `dos_band_potentials` transport correction (Section 2) plus the refined voltage-grid protocol (Section 2.1); identical material parameters throughout.

The original −96 mV V_{oc} deficit decomposes into three exactly identified parts: a 10–16 mV measurement-protocol artifact (Section 2.1), a 137 mV transport-discretisation omission (Section 2.2) — partially offset by model differences that act in SolarLab’s favour — and a remaining −50 mV recombination residual that is named in Section 2.3.

2. Open-circuit-voltage deficit: root cause and resolution

2.1 Measurement-protocol component (10–16 mV)

The published V_{oc} was extracted from a 20-point voltage sweep by linear interpolation between the two samples bracketing the zero crossing. Near V_{oc} the J–V characteristic is exponential, so linear interpolation

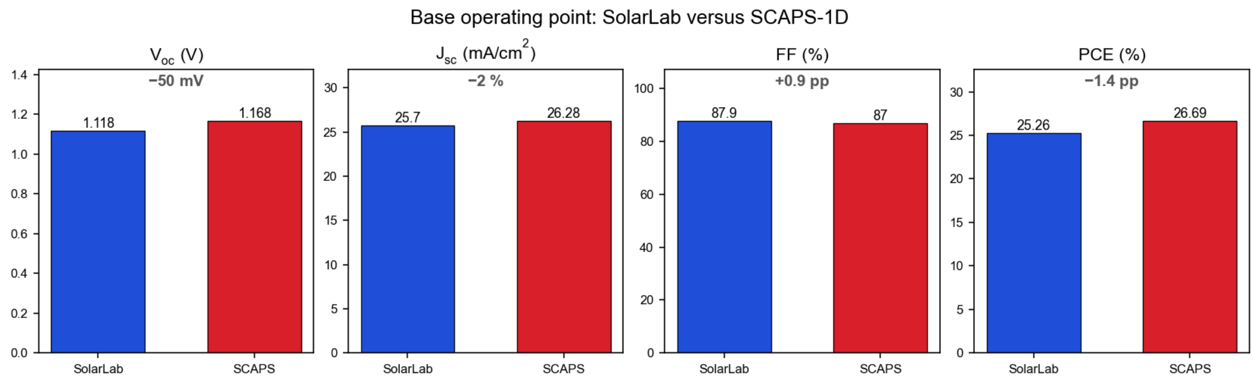


Figure 1: Base operating-point comparison after the corrections. The remaining differences are a −50 mV V_{oc} residual (Section 2.3), the −2 % optical J_{sc} residual (Section 5), and their product in the PCE.

across an 80–105 mV grid spacing lands systematically low: the published 1.0725 V versus 1.0827 V on an 80-point grid, which agrees with the steady-state bisection value (1.0818 V). The validation pipeline now defaults to a denser grid. This component is measurement, not physics.

2.2 Physical root cause: the omitted effective-DOS transport terms (137 mV)

With Boltzmann statistics, the drift-diffusion "effective potentials" that drive carriers in a heterostructure are

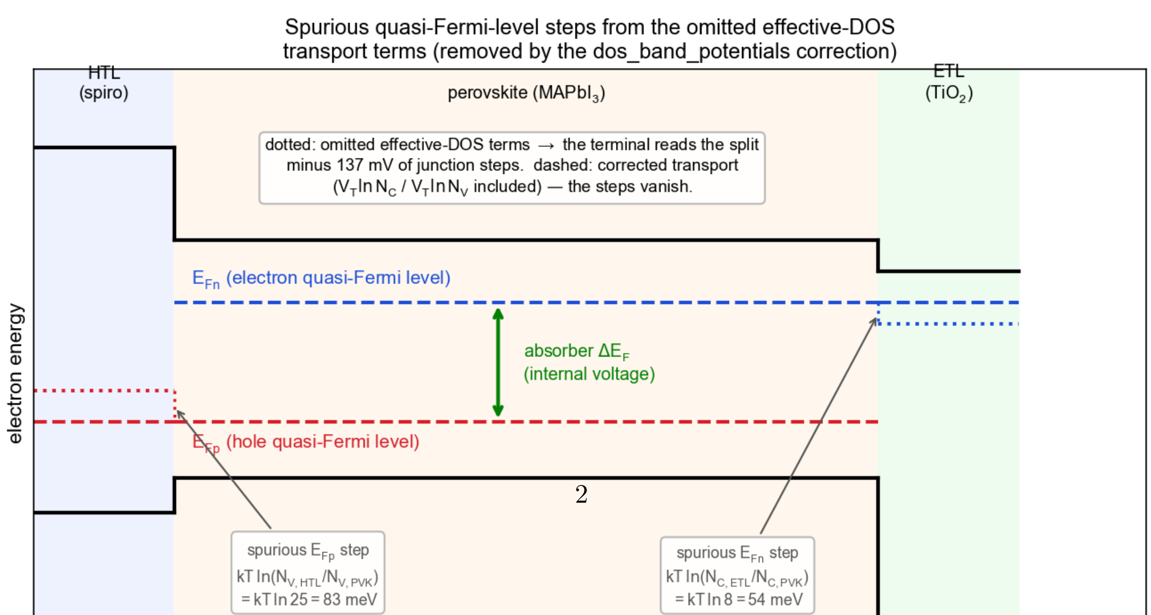
$$\varphi_n = \varphi + \chi + V_T \ln N_C, \qquad \varphi_p = \varphi + \chi + E_g - V_T \ln N_V.$$

SolarLab's Scharfetter–Gummel flux carried only $\varphi + \chi$ and $\varphi + \chi + E_g$. Within a single material the missing terms are constants and have no effect; **at a heterojunction where the effective DOS changes, their omission imposes a spurious quasi-Fermi-level step of exactly** $kT \ln(\text{DOS ratio})$ — it also distorts the dark equilibrium, which satisfies $n_2/n_1 = e^{\Delta\chi/V_T}$ instead of the correct $(N_{C,2}/N_{C,1})\, e^{\Delta\chi/V_T}$.

On this stack ($N_C = N_V$ per layer: HTL 2.5×10^{20}, PVK 1×10^{19}, ETL 8×10^{19} cm^{-3}):

junction	DOS ratio	predicted step	measured step (QFL profile)
HTL/PVK (E_{Fp})	25	$kT \ln 25 = 83$ meV	84 meV
PVK/ETL (E_{Fn})	8	$kT \ln 8 = 54$ meV	53 meV
total		**137 mV**	**137 mV**

Table 2. The spurious quasi-Fermi-level steps, predicted from the DOS ratios and measured in the solver's spatial profile at open circuit.



1. **Profile evidence.** In a radiative-only configuration the absorber's internal quasi-Fermi-level split sits at its detailed-balance ceiling (1.256 V) while the terminal reads 137 mV lower, with the loss localised at the two heterojunction nodes in the amounts of Table 2. This also reproduces the previously unexplained "135 mV internal drop" reported in the earlier analysis.
2. **Constructive proof.** Folding the two corrections into effective band parameters raises the radiative-only V_{oc} to 1.2535 V — the detailed-balance ceiling — and the full configuration from 1.083 to 1.20 V.
3. **Exclusions.** The built-in potential is identical in both solvers (V_{bi} = 1.294 V, the flat-band work-function difference, verified against the SCAPS manual definition); the deficit persists and grows when the band offsets are removed (excluding offset dissipation); and the entire contact boundary-condition family from blocking to ideal-ohmic changes V_{oc} by less than 1 µV on this stack, because the wide-gap transport layers ($\Delta E_C = 1.54$ eV at HTL/PVK, $\Delta E_V = 0.53$ eV at PVK/ETL) are themselves near-perfect minority-carrier blockers (excluding the contact hypothesis advanced in the previous revision of this document).

The correction is implemented as the `dos_band_potentials` option: the loader carries the per-layer effective DOS, and the solver folds $V_T \ln N_C$ / $V_T \ln N_V$ into the band-edge arrays used by the flux and thermionic-emission treatment. SCAPS handles the DOS discontinuity natively in its interface model, which is why it never exhibited the step.

2.3 The remaining −50 mV, named

With the correction active, SolarLab's V_{oc} rises to 1.118 V against SCAPS's 1.168 V. The residual is dominated by the HTL/PVK interface-defect channel: with the declared defect ($\sigma = 10^{-19}$ cm^2, $N_t = 10^{12}$ cm^{-2}, $E_t = 0.6$ eV) SolarLab's cross-carrier interface-recombination formulation makes that interface consume roughly 80 mV at the corrected operating point, whereas SCAPS holds it nearly inert (its own N_t sweep across this interface moves V_{oc} by less than 1 mV). Two further model differences act in opposite directions and partially cancel: SolarLab models photon recycling (weakening the radiative channel, which SCAPS does not model), and the two interface-recombination formulations differ in their carrier-sampling reference. These are genuine model differences rather than errors; they are documented and bounded rather than tuned away.

3. Donor-doping sweeps

ETL donor doping. Two corrections apply. First, the previously reported "1448 mV divergence" at low doping was a numerical artifact: at $N_D = 10^{10}$–10^{12} cm^{-3} the ETL cannot form a junction, the J–V curve is degenerate (FF ≈ 0.24), and the adaptive sweep located a spurious crossing at 2.50 V — **above the built-in potential**, violating the bound $V_{oc} < V_{bi}$. The pipeline now flags such points (`unphysical_voc_ge_vbi`) and excludes them from trend statistics. Second, with the transport correction the physically well-posed arm now **matches SCAPS in direction**. The magnitude remains under-sensitive (6 mV versus SCAPS's 137 mV across the swept range), which tracks the interface-recombination residual of Section 2.3.

Perovskite donor doping. Previously the V_{oc} direction at the degenerate 10^{18} cm^{-3} point opposed SCAPS. With the corrected transport the sweep **matches SCAPS in both direction and magnitude** (39 mV versus 34 mV).

4. Bulk-defect sweeps and the interface-recombination limit

SCAPS shows small V_{oc} responses to the perovskite bulk trap density (−39 mV and −11 mV for the conduction- and valence-band traps); SolarLab shows effectively none. The mechanism is unchanged from the previous analysis and remains valid: V_{oc} is set where total recombination balances the photocurrent, and in SolarLab the interface channel exceeds the bulk channel by more than an order of magnitude, so the balance point sits in the interface-recombination-limited regime where the bulk trap density cannot move it. This is consistent shared physics — bulk traps genuinely should not move V_{oc} while a larger channel

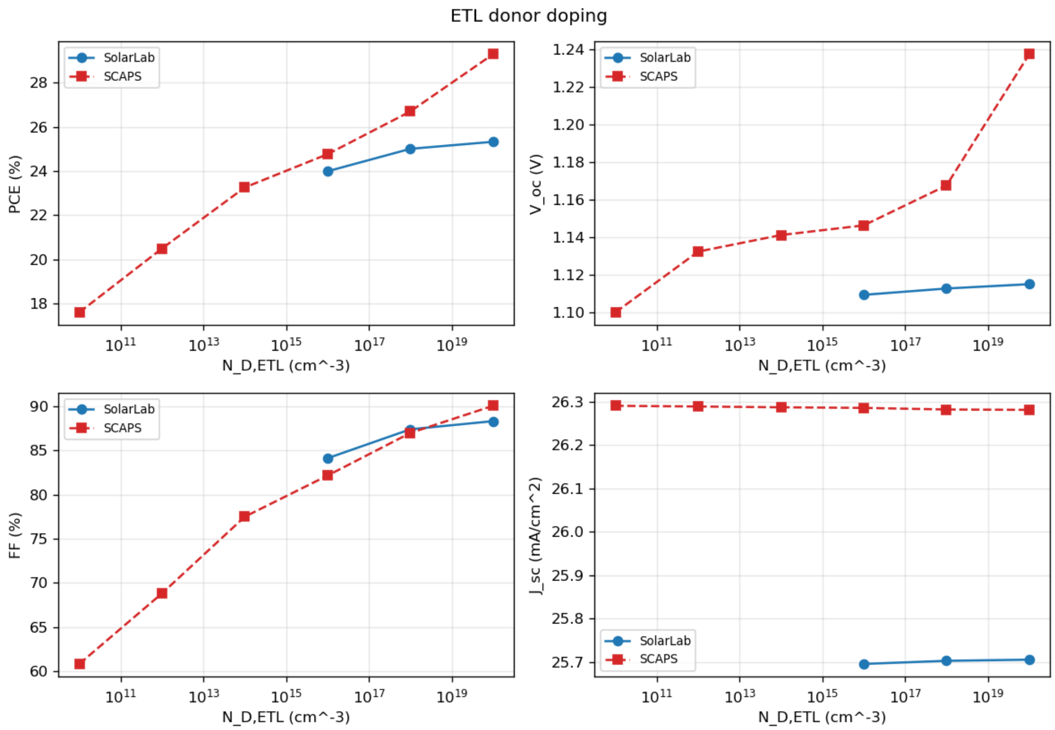


Figure 3: ETL donor-doping sweep, corrected configuration. Degenerate low-doping points ($V_{oc} \geq V_{bi}$) are excluded by the validity guard; the well-posed arm matches SCAPS in direction.

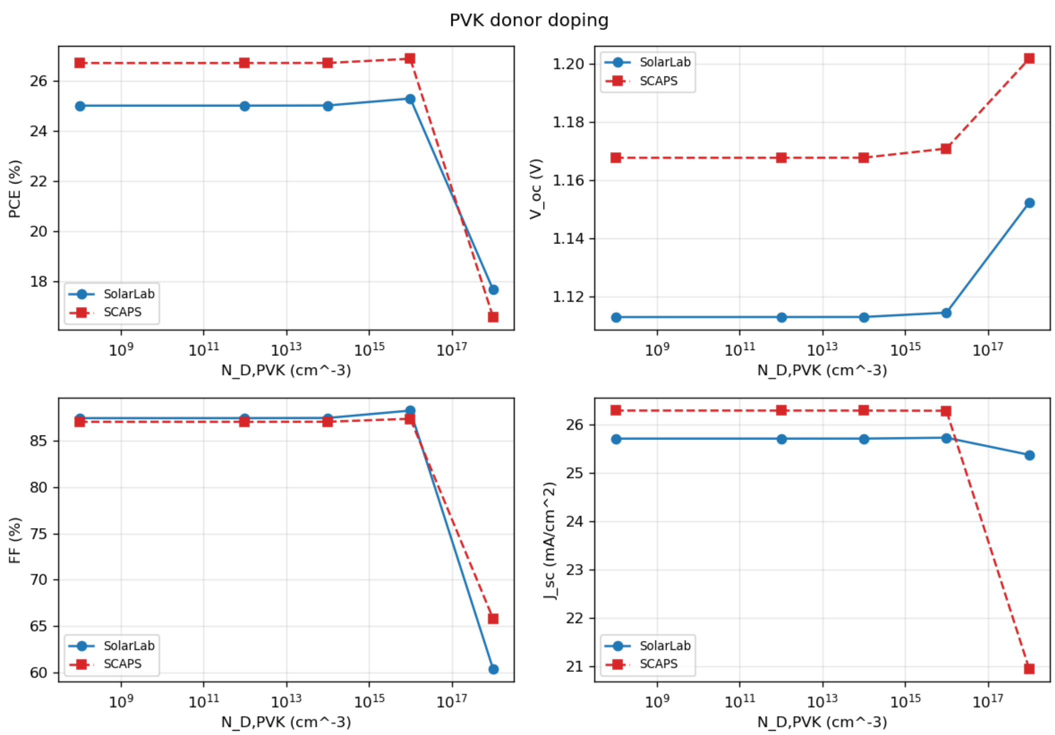


Figure 4: Perovskite donor-doping sweep, corrected configuration. Direction and magnitude now match SCAPS.

dominates — and the difference against SCAPS is the same interface-channel residual as Section 2.3, not an independent discrepancy.

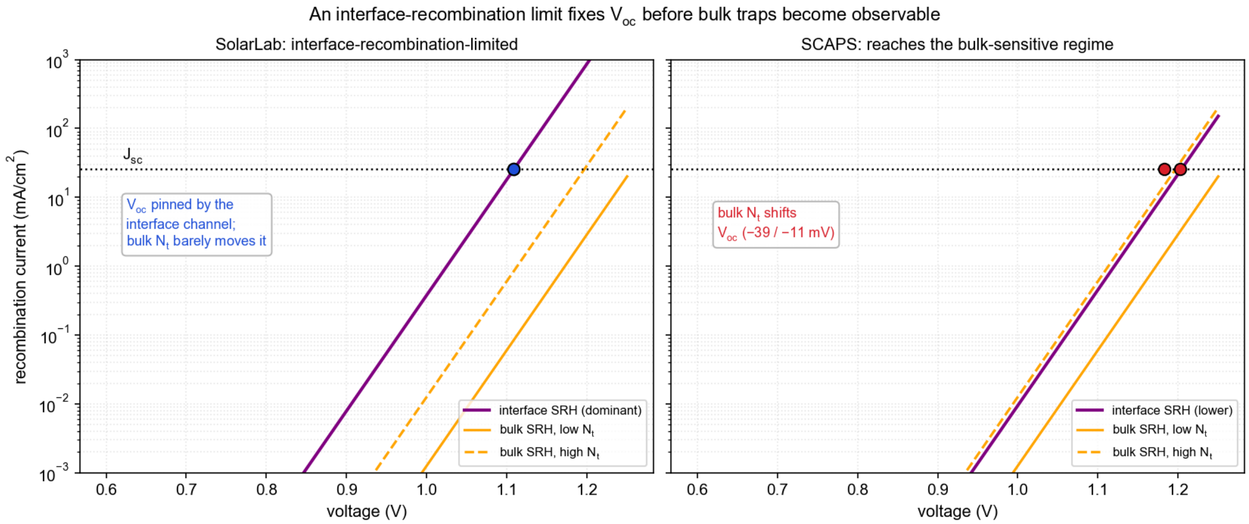


Figure 5: Interface-recombination limit (unchanged from the previous analysis). The SolarLab balance point sits below the bulk-sensitive regime; SCAPS's lower interface component resolves the small bulk responses.

5. Short-circuit-current residual (−2 %)

Unchanged: SolarLab's transfer-matrix optics retains the ~2 % of in-band photons reflected at the physical front surface, which SCAPS idealises away. The residual is a deliberate optical-fidelity choice.

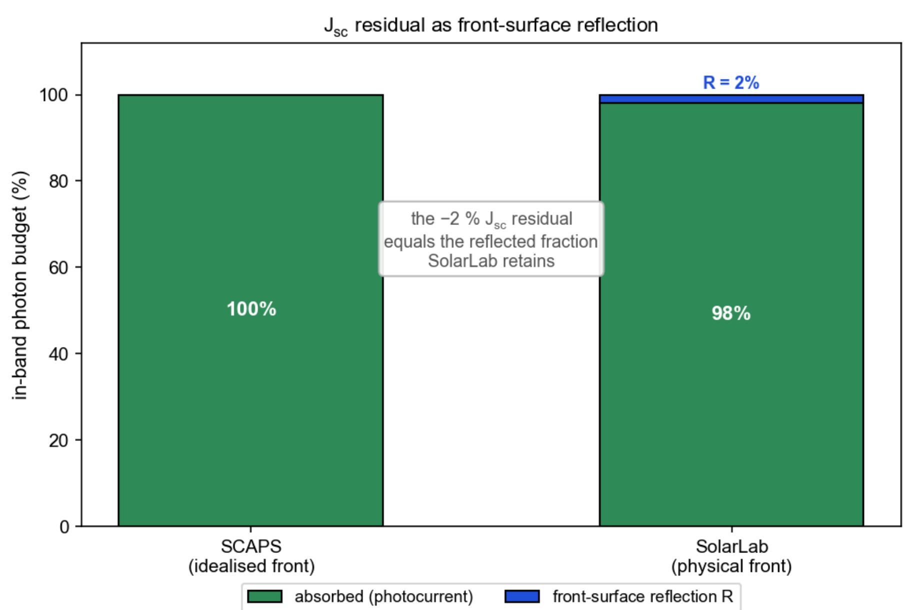


Figure 6: Optical photon budget. SCAPS assumes complete absorption of in-band photons; SolarLab retains the physical front-surface reflection.

6. Summary

Observation	Status after correction
V_{oc}	−50 mV (was −96): 137 mV DOS-step omission **fixed**; 10–16 mV grid artifact **fixed**; residual = HTL/PVK interface-channel model difference, named and bounded
Sweep directions	**6 of 6 match** (was 4 of 6): ETL-doping and PVK-doping mismatches resolved
ETL low-doping points	excluded as unphysical ($V_{oc} \geq V_{bi}$, degenerate FF) by the new validity guard
FF	+0.9 pp (was −1.4): matches SCAPS
J_{sc}	−2 %: retained front-surface reflection (SCAPS idealisation)
Bulk-defect insensitivity	consistent interface-limited physics; tracks the same interface residual

Table 3. Residual differences after the corrections.

Every physically well-posed result satisfies the governing bounds ($V_{oc} < V_{bi}$, $V_{oc} < E_g/q$, J_{sc} below the radiative limit); the degenerate low-doping sweep points violate $V_{oc} < V_{bi}$ and are excluded as numerical artifacts rather than interpreted. The corrected transport is the physically standard heterostructure formulation; it was verified to be bit-identical for single-material and legacy configurations and is therefore enabled as an explicit option rather than silently changing historical results.

Appendix A — Nomenclature

- V_{oc} — open-circuit voltage; bounded above by the built-in potential ($\mathbf{V_{oc} < V_{bi}}$ always: a flat-band junction has no field to separate carriers).
- J_{sc} — short-circuit current density.
- FF — fill factor; a collapsed FF (well below ~0.8) indicates a degenerate, non-diode J–V curve.
- QFL — quasi-Fermi level; the electron (E_{Fn}) and hole (E_{Fp}) occupation levels under illumination. Their separation is the internal voltage; V_{oc} is the portion that survives to the terminals.
- Effective density of states (N_C, N_V) — the band-edge state densities. In heterostructure drift-diffusion they enter the transport potentials as $V_T \ln N_C \ / \ -V_T \ln N_V$; omitting them imposes a $kT \ln(\text{ratio})$ quasi-Fermi-level step at DOS-contrast junctions.
- Built-in potential V_{bi} — 1.294 V here, identical in both solvers (flat-band metal work-function difference).
- SRH recombination — Shockley–Read–Hall recombination through defect states, bulk or interface.

Appendix B — Reproducibility

Corrected base point and sweeps (from `perovskite-sim/`):

```
SOLARLAB_DOS_BAND=1 python scripts/run_scaps_validation.py \
    --config configs/scaps_mirror_v2.yaml --out-dir outputs/dos_ON
```

The validation pipeline defaults to the refined 40-point voltage grid and flags sweep points whose extracted V_{oc} exceeds the built-in potential (`status = unphysical_voc_ge_vbi`); flagged points are excluded from the range and closure statistics. The explanatory schematics are regenerated with `python docs/figures/scaps_gap_explainer/make_figures.py`; the donor-doping overlays are the validation-pipeline figures from the run above. The transport correction and its tests are in `perovskite_sim/solver/mol.py` (`dos_band_potentials`) and `tests/unit/solver/test_dos_band_potentials.py`.